Madhav Sinha

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Employment

Postdoctoral Scholar at Center for Quantum Phenomena, New York University — July'26 - Present

Education

Rutgers University, New Brunswick, PhD Physics — August'20 – May'26

- **GPA:** 4.0/4.0
- **PhD Thesis:** Topological defects in conformal field theories and their lattice realizations
- **PhD Advisor:** Prof. Ananda Roy

Indian Institute of Science Education and Research, Pune, BS-MS Physics — August'15 – May'20

- **GPA:** 9.1/10 - Distinction
- **Master's Thesis:** Solvable Models and Mathematical Formulation of CFT
- **Master's Thesis Advisors:** Profs. Vladimir Bazhanov and Murray Batchelor (Australian National University)

Publications

Publications can be found on Inspire

- Madhav Sinha, Thiago Silva Tavares, Hubert Saleur, and Ananda Roy. Lattice Topological Defects in Non-Unitary Conformal Field Theories. 4 2026
- Madhav Sinha, Thiago Silva Tavares, Ananda Roy, and Hubert Saleur. Integrability and lattice discretizations of all topological defect lines in minimal CFTs. *Nucl. Phys. B*, 1023:117308, 2026
- Thiago Silva Tavares, Madhav Sinha, Linnea Grans-Samuelsson, Ananda Roy, and Hubert Saleur. Integrable RG flows on topological defect lines in 2D conformal field theories. *JHEP*, 04:148, 2025
- Ranveer Kumar Singh, Madhav Sinha, and Runkai Tao. Rationality of Lorentzian lattice CFTs and the associated modular tensor category. *Letters in Mathematical Physics*, 115(134), 2025
- Ranveer Kumar Singh and Madhav Sinha. Non-Chiral Vertex Operator Algebra Associated To Lorentzian Lattices And Narain CFTs. *SciPost Phys.*, 17:047, 2024
- Madhav Sinha, Fei Yan, Linnea Grans-Samuelsson, Ananda Roy, and Hubert Saleur. Lattice realizations of topological defects in the critical (1+1)-d three-state Potts model. *JHEP*, 07:225, 2024

Seminar Talks

Center for Quantum Phenomena Seminar at New York University	October '25
Condensed Matter Theory Seminar at Leibniz University Hannover	October '24
CMP/QFT Seminar at Institute for Advanced Study	April '24
Integrable Systems Seminar at University of Leeds	November '23

Conferences and Workshops

Student Workshop on Integrability - Ljubljana	*Talk* - June '26
Les Houches Summer School 2025: Exact Solvability and Quantum Information	*Short Talk* - August '25
ICTP - SAIFR - Program on Anomalies, Topology and Quantum Information in Field Theory and Condensed Matter Physics	*Poster* - June '25
KITP - Generalized Symmetries in Quantum Field Theory	April '25
Autumn school on Quantum Integrable Models at DESY, Hamburg	October '24
Prospects in Theoretical Physics (PiTP) at Institute for Advanced Study	*Poster* - July '24

Rutgers Center for Materials Theory Fall 2023 Symposium	*Poster* - September ’23
Young Researchers Integrability School and Workshop (YRISW), Durham	*Poster* - July ’23
Workshop on Conformal Bootstrap at GGI - Florence	October’22
New Jersey Quantum Matter and Information (NJQMI) Forum at Princeton	April ’22
Boundaries and Defects in CFT and Holography held at Princeton	March ’22
Baxter 2020 at Australian National University	February ’20
Workshop on Volume Conjecture and related topics in Knot Theory at IISER Pune	December ’18

Awards and Fellowships

Torrey Fellowship, Rutgers Physics and Astronomy Department	September ’24 - January ’25
Rutgers School of Arts and Sciences Excellence Fellowship	September ’24 - May ’25
Australian National University - Future Research Talent Award	May ’19 - March ’20
University of Göttingen - Erasmus	May ’18 - Aug ’18
Indian Academy of Sciences’ Summer Research Fellowship Programme	May - July ’17
Kishore Vaigyanik Protsahan Yojana (KVPY) scholarship which supports undergraduates in science in India	August ’15 - July ’20
National Talent Science Examination, scholarship by the Indian government for high school students	April ’13 - May ’15

Relevant Coursework

QFT 1 & 2, Differential Topology, Vertex Operator Algebra, Many Body Physics, Quantum Computing, Frobenius Algebra and Topological Field Theory, Applied Group Theory, General Relativity	Rutgers, New Brunswick
Introductory QFT, Algebraic Topology, Differential Geometry, Differential Equations, Atomic and Molecular Physics, Nuclear Physics, Physics Labs	IISER Pune

Teaching Experience

Rutgers University

Teaching Assistant for General Physics Lab	Spring ’24
Grader for Quantum Computation	Fall ’23
Conducted Recitations and grader for Honors Physics I: Classical Mechanics	Fall ’22
Teaching Assistant for Extended Analytical Physics	Spring ’22
Teaching Assistant for Analytical physics	Fall ’21
Teaching Assistant for General Physics Lab (Online)	Spring ’21

Technologies

Programming Languages: Python, Mathematica, Fortran
Markup Languages LaTeX